

Airspeeds

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1 CAS to TAS

$$P_0 = 101325, a_0 = 340.29$$

$$q_c = P_0 \left(\left(\frac{1}{5} \left(\frac{CAS}{a_0} \right)^2 + 1 \right)^{\frac{7}{2}} - 1 \right)$$

$$M = \sqrt{5 \left[\left(\frac{q_c}{P_{si}} + 1 \right)^{\frac{2}{7}} - 1 \right]}$$

$$TAS = M \cdot a_{si}$$

$$P_{si} = P_0 \cdot \frac{288.15}{288.15 + L \cdot H}^{\frac{gM_0}{R^* \cdot L}}$$

$$L = -\frac{6.5}{1000}, g = 9.80665, M_0 = 28.9644, R^* = 8.31432 \times 10^3$$

$$a_{si} = \sqrt{\gamma RT}, \quad \gamma = 1.4 \quad R = 287.053$$

2 TAS to CAS

$$M = \frac{TAS}{a_{si}}$$

$$q_c = P_{si} \left[(1 + 0.2M^2)^{\frac{7}{2}} - 1 \right]$$

$$CAS = a_0 \sqrt{5 \left[\left(\frac{q_c}{P_0} + 1 \right)^{\frac{2}{7}} - 1 \right]}$$

3 TAS Correction

The TAS generated from the above equations differ from the TAS generated from the whiz wheel in a predictable fashion. I ran three regressions of the residuals $r_{TAS} = \widehat{TAS} - TAS$, where $\widehat{TAS}$ is my TAS from the equations and TAS is the whiz wheel generated answers, with $\widehat{TAS}$ and found a quadratic to be best. The error seems to be relatively invariant to input altitude and OAT, simply changing in magnitude with the TAS. This suggests some compressibility correction being made in one of the methodologies.

Table 1: IV Analysis: Investment on Bribes table

	r_{TAS}		
	(1)	(2)	(3)
$\widehat{TAS}$	0.068*** (0.003)	-0.048*** (0.005)	0.01 (0.02)
$\widehat{TAS}^2$		$1.68 \times 10^{-4}***$ (7×10^{-6})	-1×10^{-5} (5×10^{-5})
$\widehat{TAS}^3$			$1.6 \times 10^{-7}***$ (5×10^{-8})
Constant	-13*** (1)	4.5*** (0.8)	-1.2 (1.9)
Observations	48	48	48
R^2	0.86	0.998	0.998

Standard error in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

